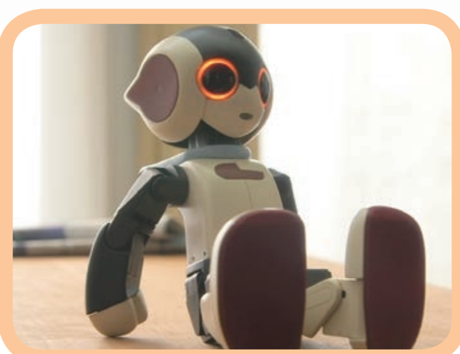
Build your own

Robi

Pack 04



Stages 11-14:
Begin assembling
Robi's body

DeAGOSTINI MODEL SPACE

Build your own Robi

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DeAGOSTINI MODEL SPACE™

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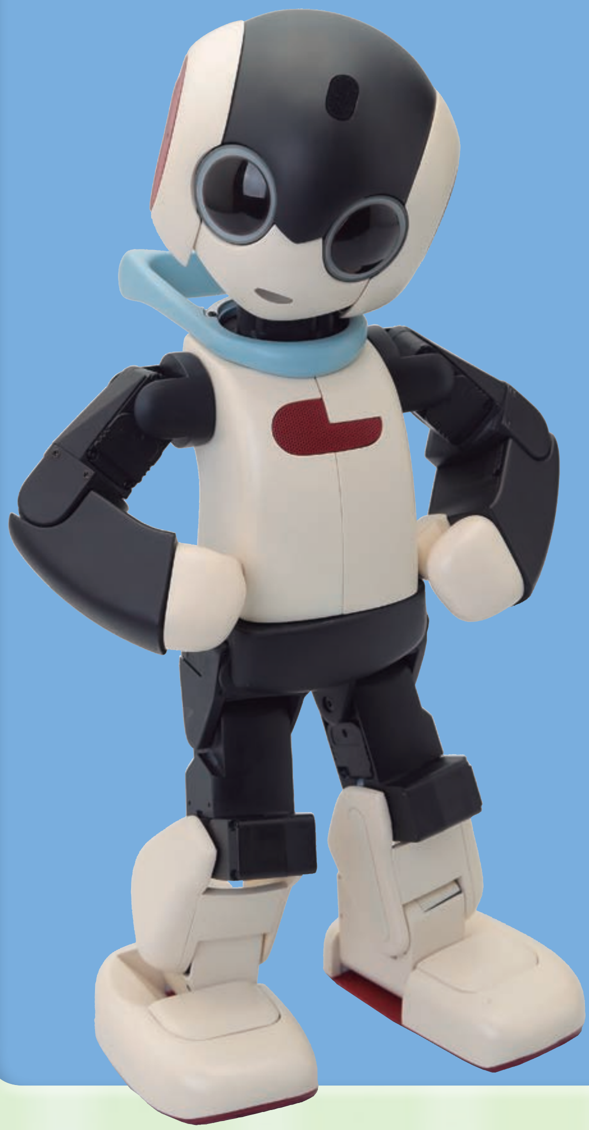


STAGE 11: ADD ANOTHER SERVO TO ROBI'S RIGHT ARM

In this assembly you will test the next servo to move Robi's arm, before fitting it to the existing right arm assembly built over the previous stages.

This stage sees you working with servo motors again. The servo supplied with this stage is ID=17, as listed on the diagram in Stage 8, and will generate the up and down motion for Robi's right arm, replicating the work of a human's shoulder. Before you fit the servo, you will test its operation and set its ID. As ever, take care when handling electronic parts!

1

PARTS TO BE ASSEMBLED

YOUR PARTS

1 Servo motor

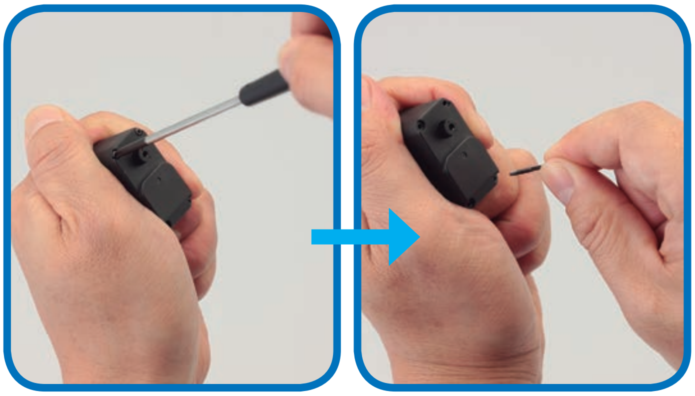
SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

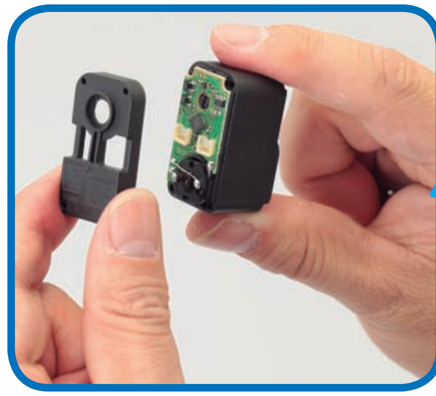
Right arm assembly and servo cable (Stage 10)

Neck stand (Stage 8)

THE SERVO AND CABLE



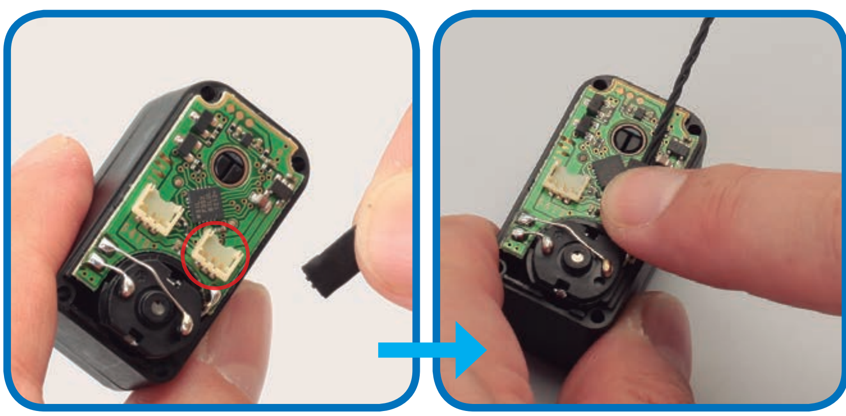
1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.



2 Gently lift the cover away, taking care not to force it or damage the circuit board underneath.

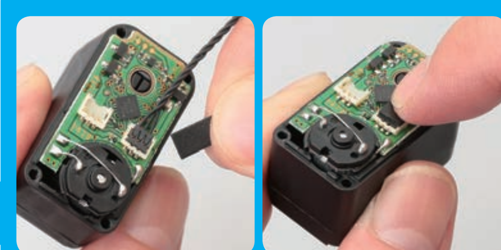
TIP!

It is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure to hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.

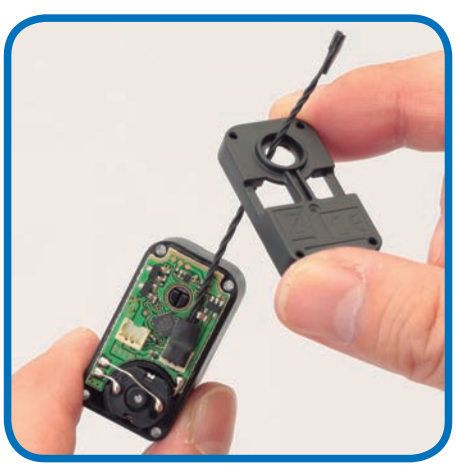


3 Hold the open servo with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous stage, and press one of its connectors into the socket circled in red, as you have done before. It does not matter which end of the servo cable you use.

TIP!



This servo cable had protective pads fitted to its connectors in the previous stage, but if you find it difficult to attach with the pad in place, it is possible to apply the pad afterwards. To do this, very carefully fit the cable's connector into the servo's socket, then remove a pad from the backing sheet and place on top of the connector and press down to seal the part.

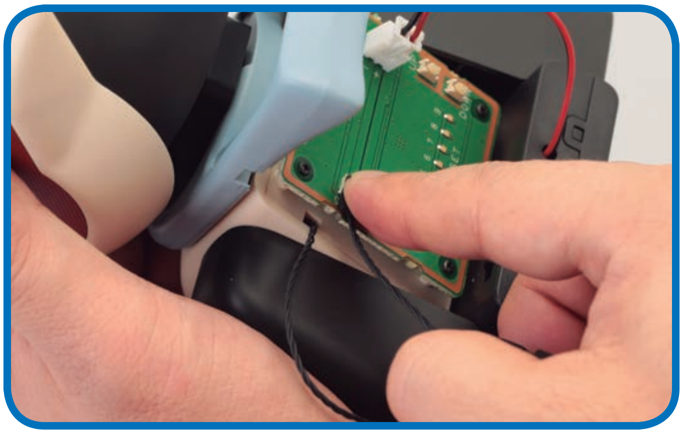


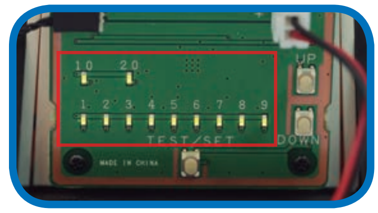
4 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

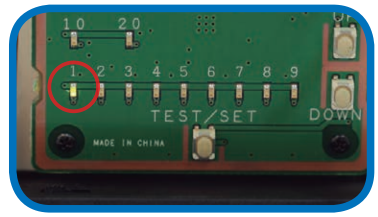
5 Press the cover back into place on the servo's main housing.

6 Use two of the screws removed in Step 1 to fasten the cover. Only tighten these lightly, as you will be removing the cover again.

TESTING THE SERVO

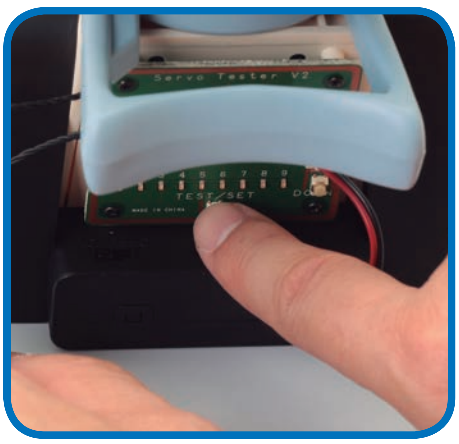


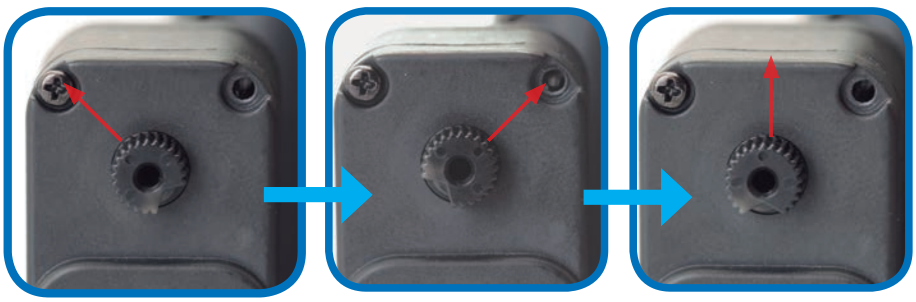




7 Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board and attach the servo cable connected to the servo in Step 3.

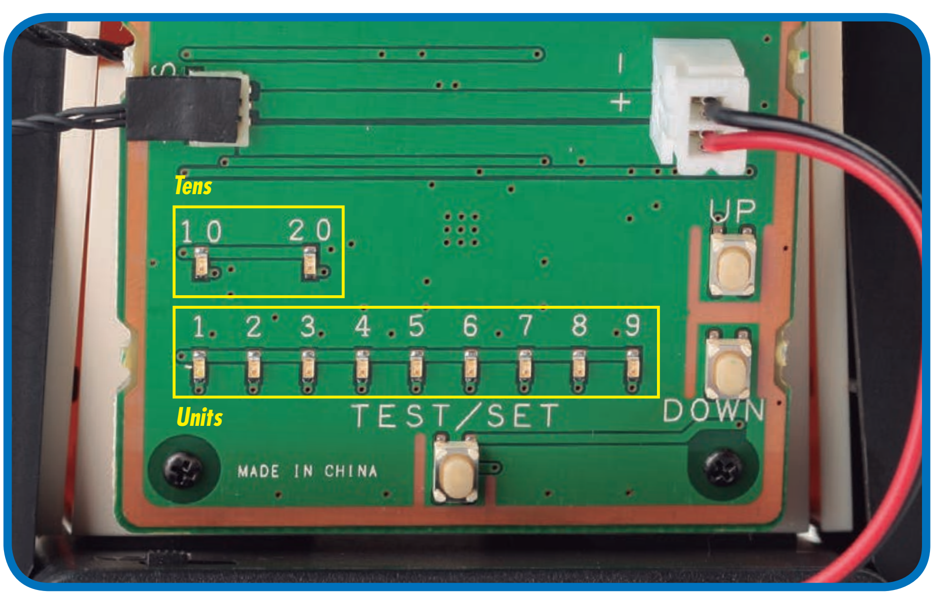
8 With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.





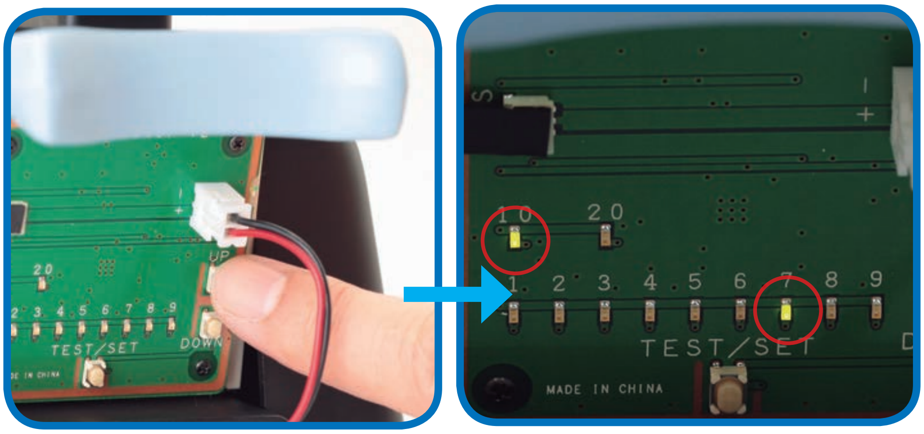
9 To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45 degrees to the left, stop, then move back across to 45 degrees to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, refer to the HELP box in Stage 8.

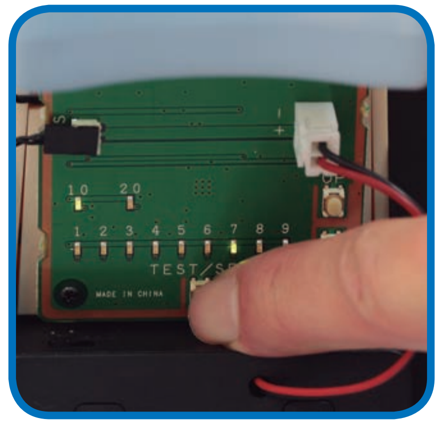
USING THE TEST BOARD



10 Re-familiarise yourself with the lights and switches on the test board, which were explained in the Assembly Guide of Stage 8. These are used to set the ID number of the servos. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units.
For example, to set the ID to 17, as you will do in this stage, you press UP 16 times. This will light up the 10 LED and the 7 LED. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.

SETTING THE ID NUMBER

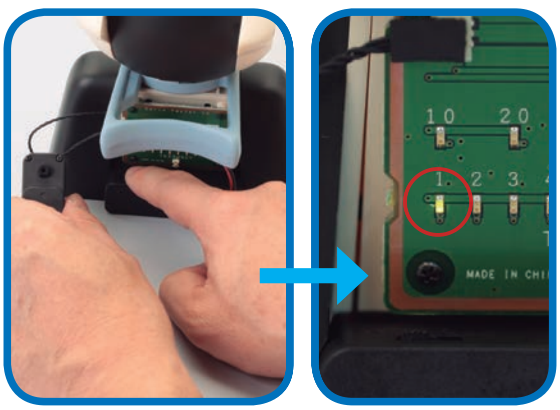


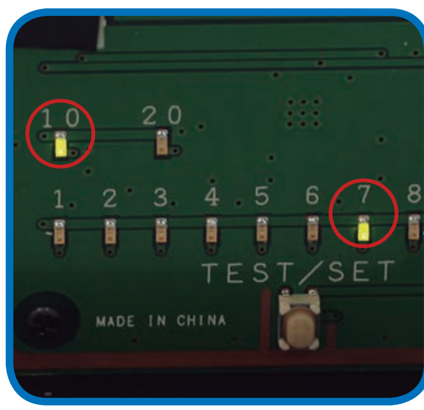


11 To set the ID number of the right upper arm servo to 17, press the UP button 16 times. Confirm the setting by checking that the 10 and 7 LEDs are lit, using the UP and DOWN buttons to correct the number if necessary.

12 Confirm the setting by pressing the TEST/SET button. The 10 and 7 lights should flash, then go out.

CHECKING THE ID NUMBER



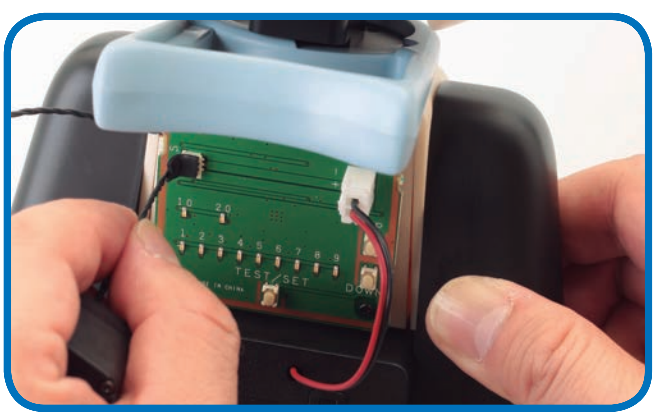


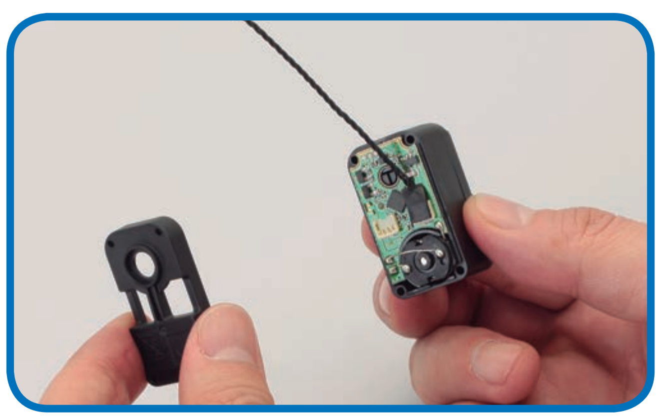
HELP!

It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake or set the wrong ID number, you can reset it by going through the setting procedure again.

13 Leaving the servo connected, switch the power OFF, then back ON. Only the 1 LED should light up.

14 To check the ID number, press TEST/SET. Now the 10 and 7 should light up. Note: It is crucial that all of Robi's servos have the correct ID numbers, so check thoroughly that you have set the desired number.

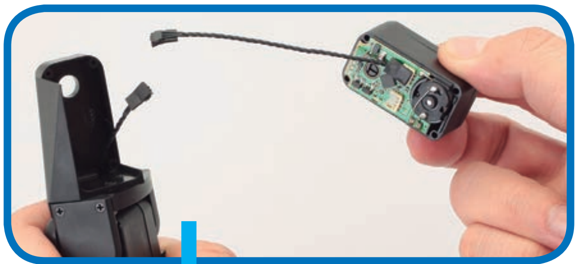




15 Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit to Robi's arm.

16 Remove the two screws holding the cover of the servo in place – keep the screws safely, but you will not need the cover again.

CONNECTING THE SERVOS

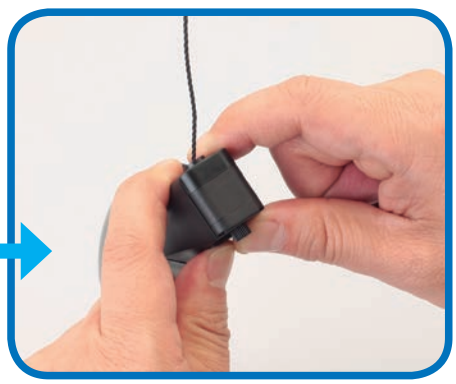


17 Fetch the right arm assembly built over the previous stages. Feed the free end of the cable attached to this stage's servo through the circular hole in the arm casing, as shown.

18 Holding both carefully, press the connector of the forearm servo's cable into the free socket on this stage's servo.

TIP!

Make sure that the cable's connector is securely clipped into the servo's socket before proceeding – if the connection is not in place correctly at this stage Robi's arm will not work as it should!



19 Making sure that the cable connectors are not disturbed, and that the free end of the cable is not caught in any of the joints, position the servo into the upper arm frame and press into place. Then, using the servo's original four screws, fasten the servo to the upper arm frame, using the screwdriver supplied with Pack 1.

Assembled right arm

Robi's right arm is nearly complete, ready to be joined to his body.

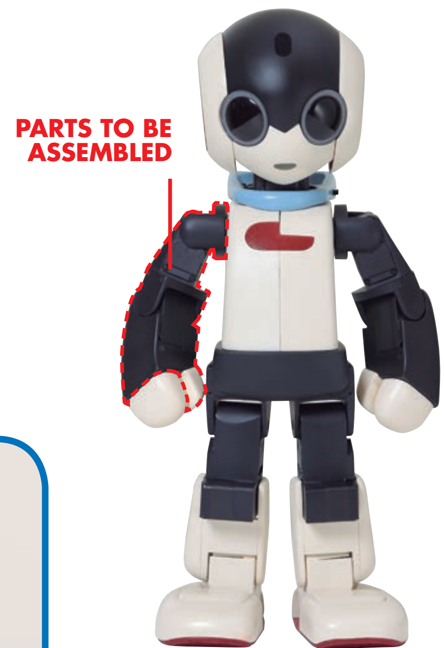
CAREFUL!

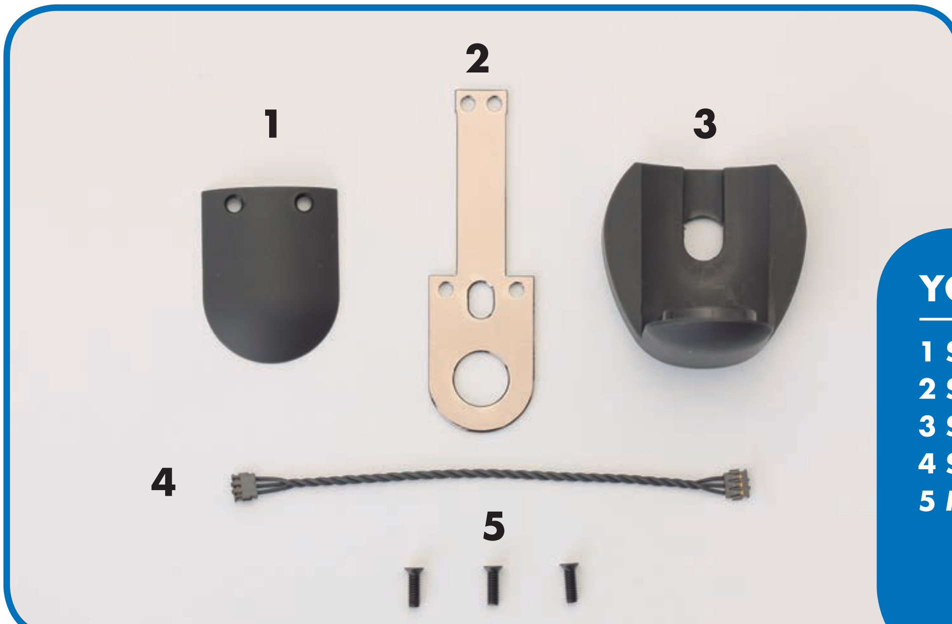
Note: for the first time, you have two of Robi's servos set with their individual ID numbers and linked with each other. Do not connect these to the test board now in any way, as this may affect the ID numbers.

STAGE 12: ADDING ROBI'S RIGHT SHOULDER JOINT

In this assembly you will fit the right arm built over the previous stages to its shoulder frame, forming the joint that will attach it to Robi's body.

By the end of this stage, Robi's right arm will be complete, and ready to be attached to his body at the shoulder over future stages. The work in this stage includes connecting servo cables to the servos in Robi's arm, so as ever care must be taken when handling these electronic parts. Similarly, be absolutely sure to follow the instructions closely, as it will be difficult to go back and change the completed assembly later.





YOUR PARTS

1 Shoulder panel
2 Shoulder joint
3 Shoulder frame
4 Servo cable (70mm long)
5 M2 x 7mm countersunk screws (x 3)

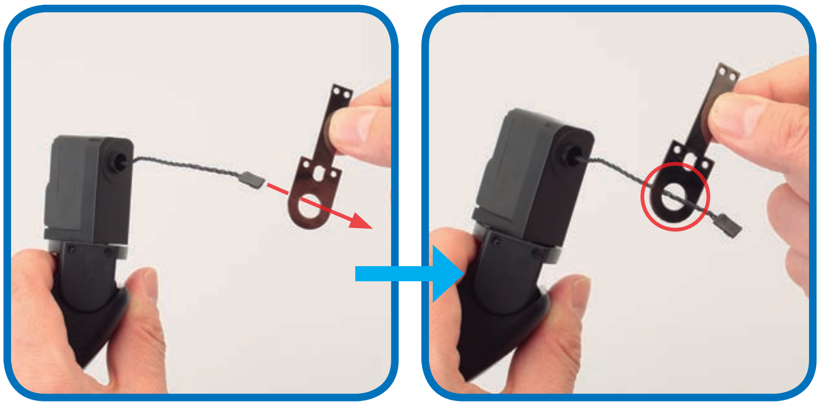
SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

Right arm assembly (Stage 11)

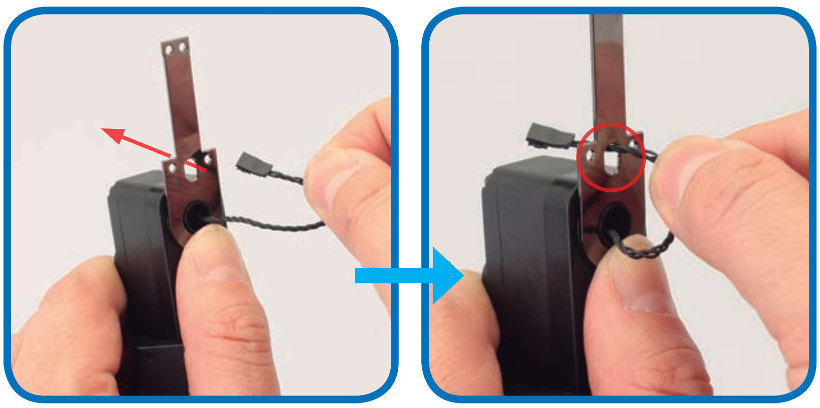
Protective pad sheet (Stage 3)

WIRING THE SHOULDER JOINT



1 Feed the servo cable coming out of the upper arm frame through the hole in the shoulder joint.

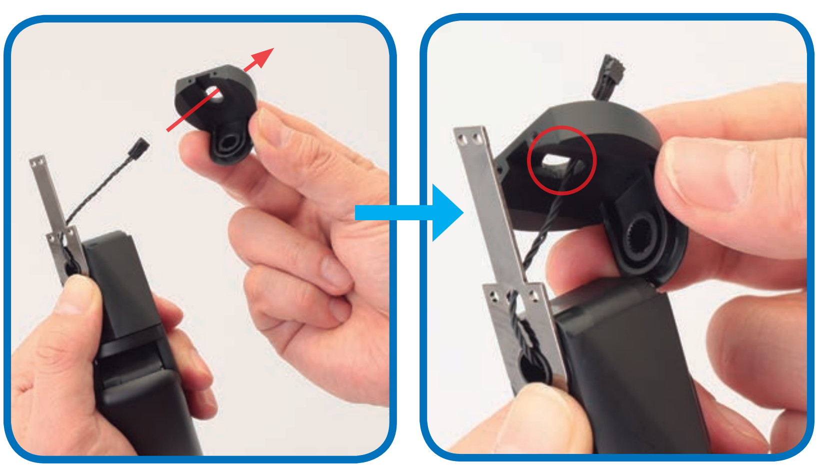
2 Run the shoulder joint along the cable and fit its circular hole around the raised opening of the upper arm frame.



3 Holding the parts in place, feed the tip of the servo cable over and through the indicated hole in the shoulder joint.

4 Keep pressing the parts together with your finger and thumb, then pull the cable gently so that it is taut.

ADDING THE SHOULDER FRAME



5 Feed the servo cable through the indicated hole in the shoulder frame and bring the parts together.

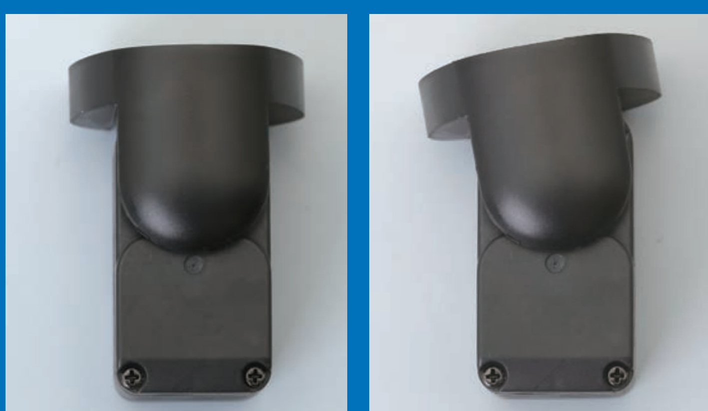
6 Join the parts by placing the 'D-cut' hole in the frame over the corresponding servo shaft (red arrow). See the Close Up section on the next page for more information on this.

THE 'D-CUT'

The round shafts of Robi's servos are moulded with a small notch in one side, creating a flat edge (resembling the letter 'D') that fits into a correspondingly shaped recess on the part to be joined. This ensures that the parts can be set correctly when in the neutral position.

The 'D-cut'

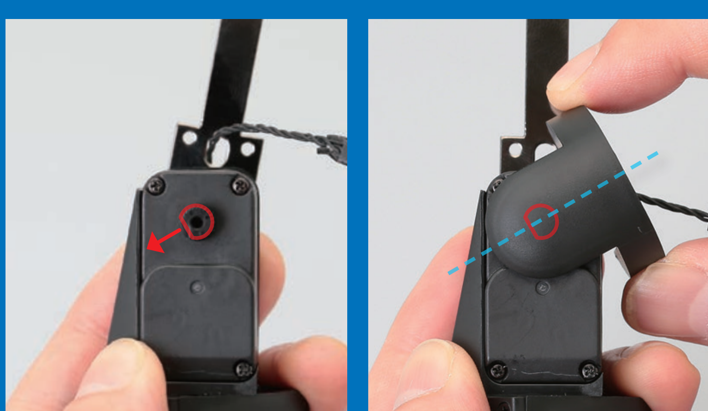
Left photo: the servo's shaft is not completely circular, but has a notch in its ribbed edge, creating a straight section of outline. Right photo: the inside of the corresponding joint mirrors these ribs, as well as the straight edge created by the shaft's notch.



Neutral position

The advantage of using a D-cut shaft and joint set-up is that it makes it very easy to assemble Robi's servos and accompanying parts correctly. When the parts are aligned correctly, as in the far left photo, they will fit together easily, while if they are not orientated as they should be, they will struggle to fit together and result in an off-centred neutral position (near left). If this happens, do not force the part, and realign it with the servo shaft until it fits easily.

Warning: applying too much force when pressing a part over the servo's shaft can result in damaging both parts!



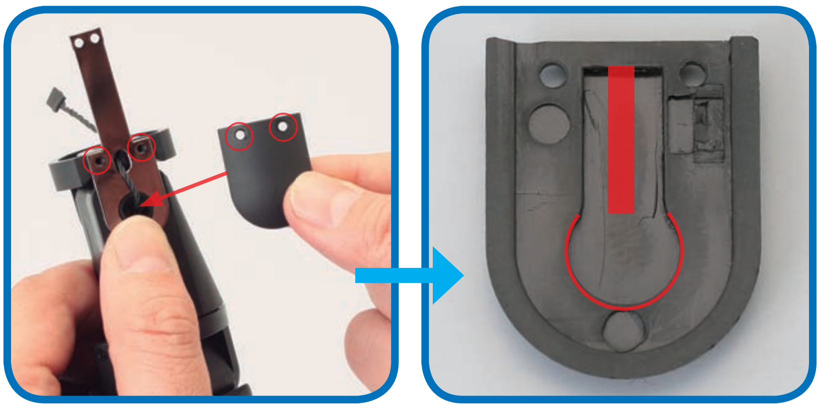
Restoring to the neutral position

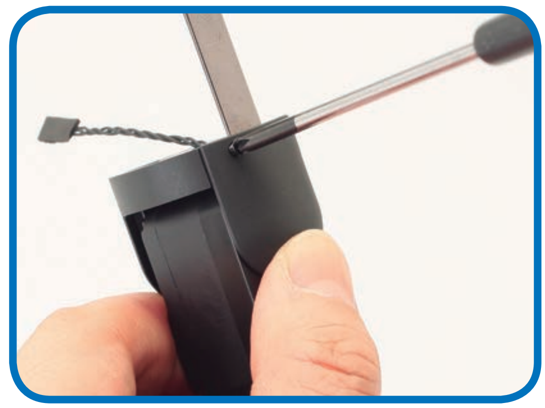
If something goes wrong when performing a test of one of Robi's servos, for example if the power is disconnected, and the servo's shaft stops turning and remains at an angle, you will need to reset the servo to its neutral position. To do this, identify the notch in the servo shaft (left photo) and place the part it will operate over it at the same angle (right photo). Then, very gently rotate the part back to the neutral position, with the servo shaft's notch facing downwards.

7 Once you have the parts aligned as they should be, press them together tightly.

8 To prevent the servo cable slipping out of position, pull it again gently so that it is taut.

SEALING THE SHOULDER

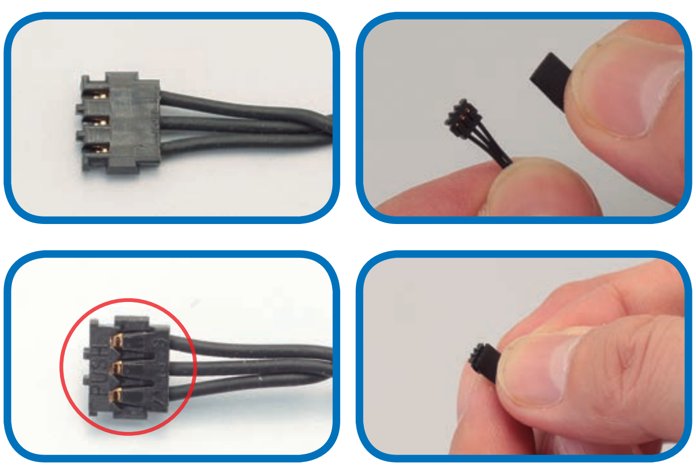




9 Place the shoulder panel over the shoulder joint, keeping the circled holes aligned. The red line in the right photo shows the groove in the shoulder panel that will take in the servo cable, while the curved line indicates the area that will accommodate the servo's shaft.

10 Once in place, secure the parts using two M2 x 7mm countersunk screws.

SEALING THE SERVO CABLE



11 As you have done previously, apply a protective pad to the side of this stage's servo cable with PUSH marked on it (circled). Repeat for both ends of the cable.

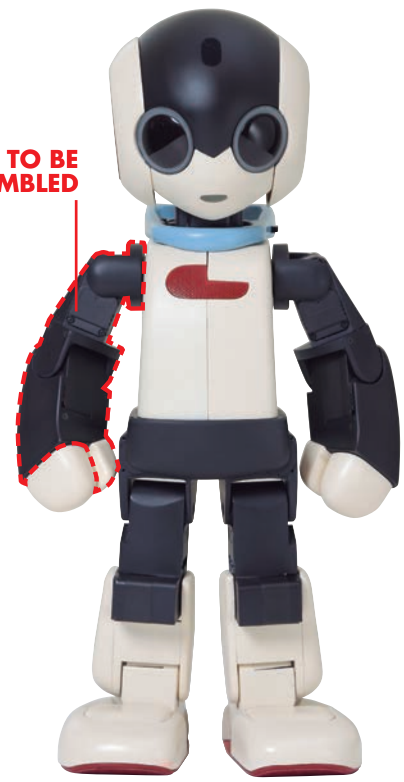
Assembled right arm and servo cable

This stage is complete. Robi's right arm is assembled and wired, ready to be joined to his body in future stages.

STAGE 13: PREPARING ROBI'S SHOULDER SERVO

In this session, you will prepare the servo that will move Robi's right shoulder back and forth, by wiring it, testing its operation then setting its ID number, ready for use.

PARTS TO BE ASSEMBLED



The servo provided today will move Robi's right arm in a linear, forward and backward, motion. As the part will be concealed inside Robi's body, it will not need to be encased in any shaped arm frames as you did for the upper and forearm servos in previous stages. To prepare this part for use, you will first test it using the test board supplied in Stage 3, and then set its ID number, which will coordinate its operation on completion.

1

YOUR PARTS

1 Servo

SAVED PARTS

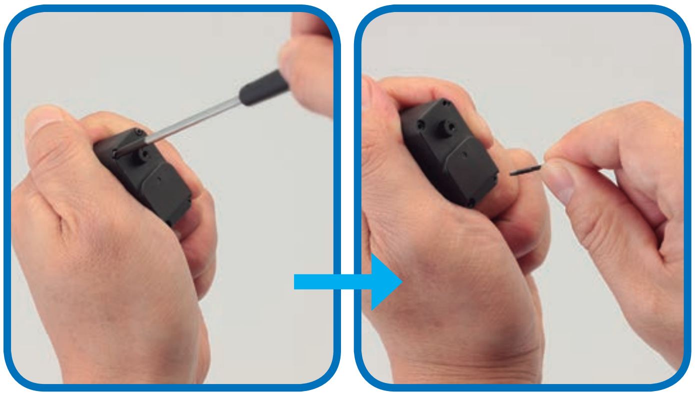
You will be using parts from the previous stages, so be sure to have these to hand before beginning.

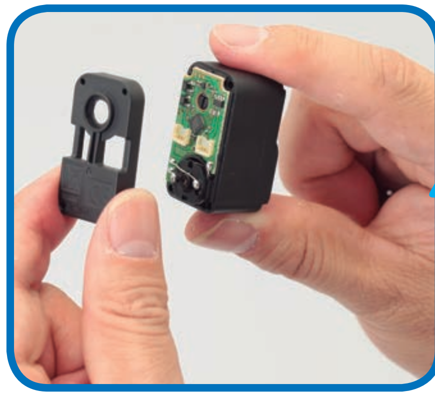
Neck stand (Stage 8)

Sealed servo cable (Stage 12)

THE SERVO AND CABLE

TIP!

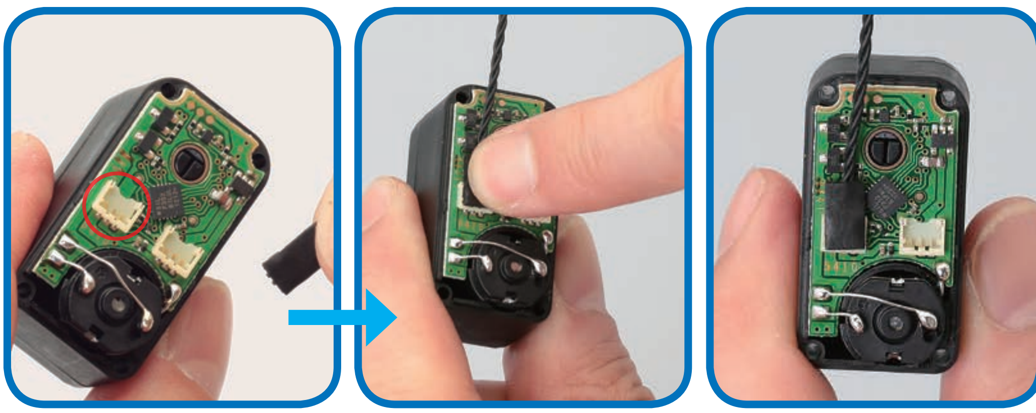




It is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Hold the open servo only by its sides, and if you put it down on your work surface during the build, do this with the open side facing upwards.

1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

2 Gently lift the cover away, taking care not to force it or damage the circuit board underneath it.



This servo cable had protective pads fitted to its connectors in the previous issue, but if you find it difficult to attach with the pad in place, it is possible to apply the pad afterwards. To do this, very carefully fit the cable's connector into the servo's socket, then remove a pad from the backing sheet and place on top of the connector and press down to seal the part.

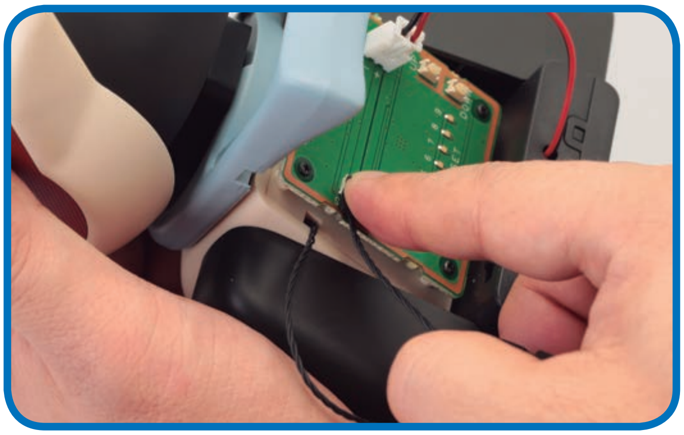
3 Hold the open servo by its sides, with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous issue, and press one of its connectors into the socket circled in red. It does not matter which end of the servo cable you use.

4 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

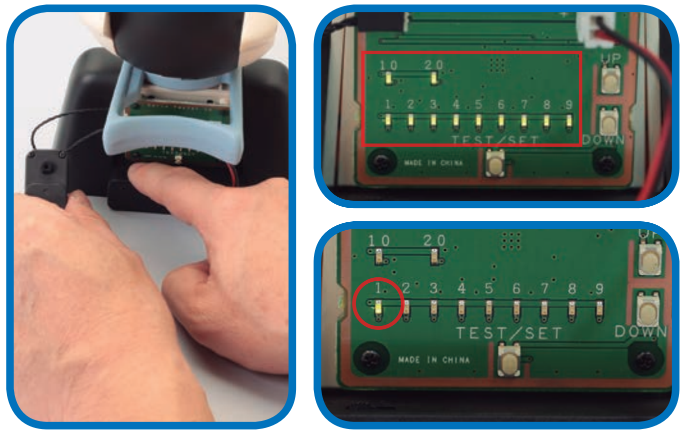
5 Press the cover back into place on the servo's main housing.

6 Use the four screws removed in Step 1 to re-attach the cover. Take care not to overtighten the screws.

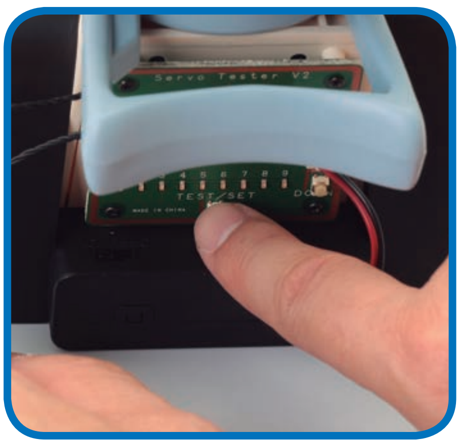
TESTING THE SERVO

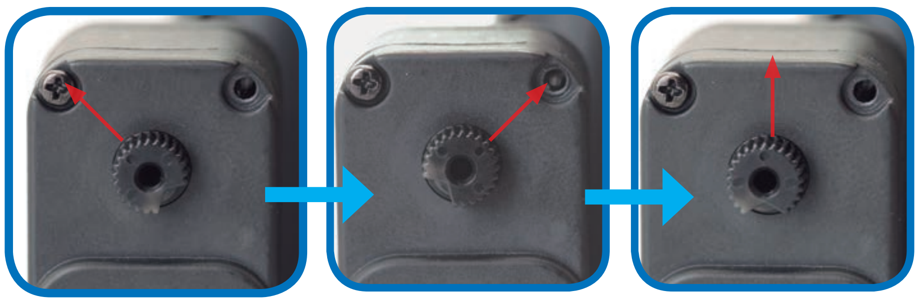


7 Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board and attach the servo cable connected to the servo in Step 3.



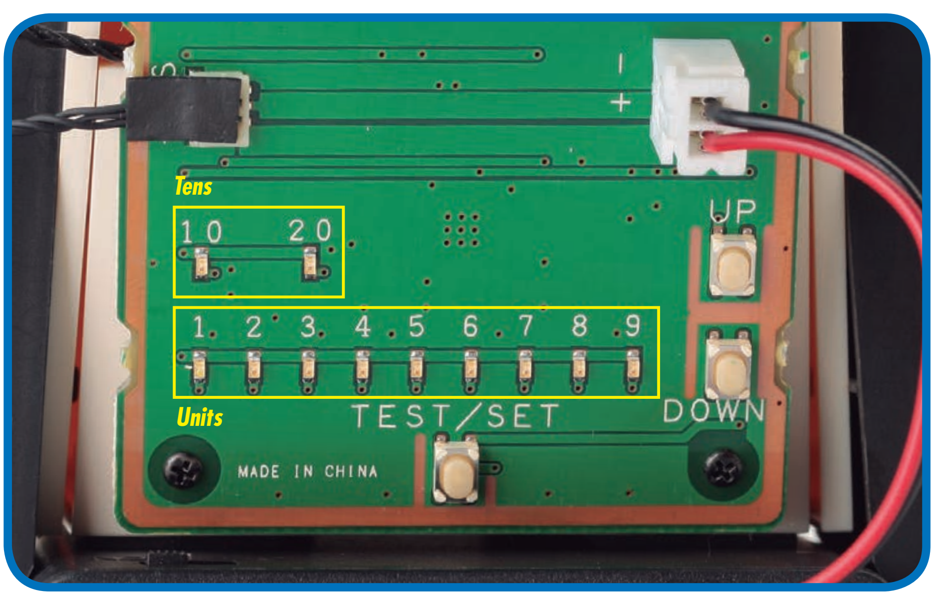
8 With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.



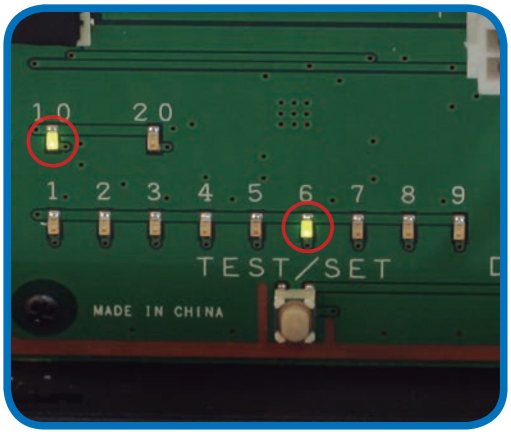


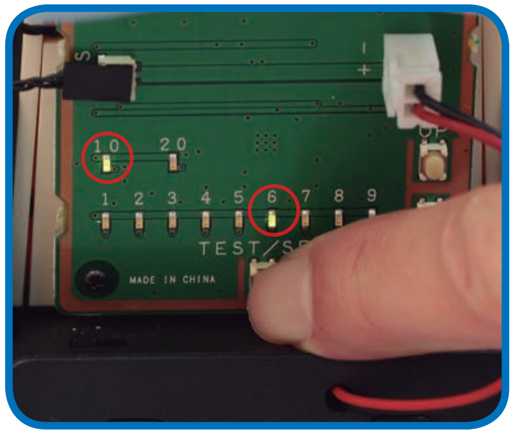
9 To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, refer to the HELP box in Stage 8.

USING THE TEST BOARD



10 In the coming stages, you will again be setting a servo's ID number using the test board, so re-familiarise yourself with the board's lights and switches which were first explained in the Assembly Guide of Stage 8. These are used to set the ID numbers of the servos. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 16, as you will do in this stage, you press UP 15 times. This will light up the 10 LED and the 6 LED. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.



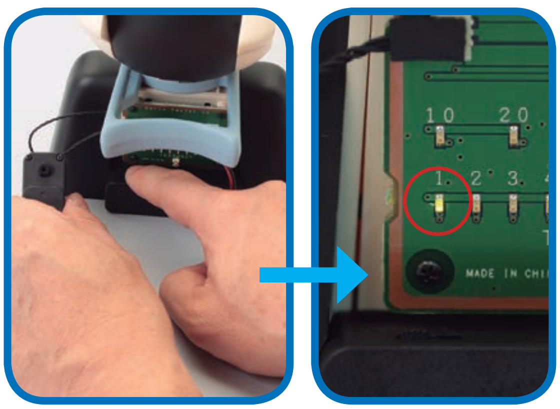


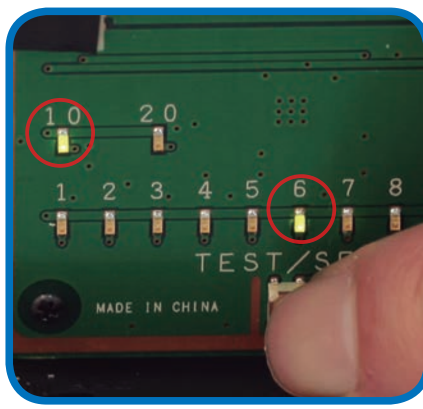
ID=16

11 To set the ID number of the right upper arm servo to 16, press the UP button 15 times. Confirm the setting by checking that the 10 and 6 LEDs are lit, using the UP and DOWN buttons to correct the number if necessary.

12 Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 10 and 6 lights should flash, then go out.

CHECKING THE ID NUMBER



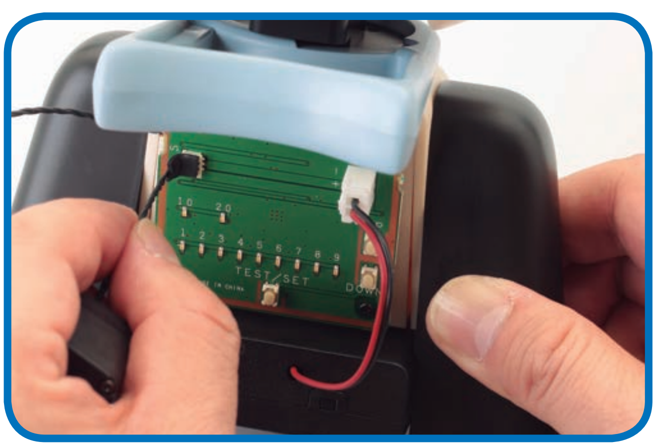


13 Leaving the servo connected, switch the power OFF, then back ON. Only the 1 LED should light up.

14 To check the ID number, press TEST/SET. Now the 10 and 6 should light up. Note: It is crucial that all of Robi's servos have the correct ID numbers, so check thoroughly that you have set the desired number.

HELP!

It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake or set the wrong ID number, you can reset it by going through the setting procedure again.



15 Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit into Robi's shoulder.

Assembled shoulder servo

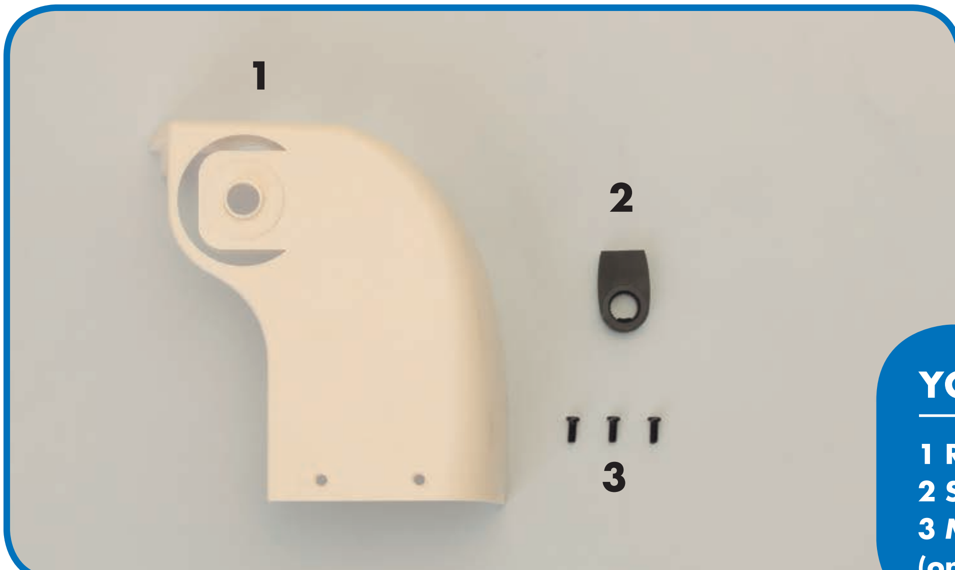
The servo that will operate Robi's shoulder is assembled, wired and set. Make sure you keep this part safely, preferably in a sealed plastic bag with its ID number (16) written on it for ease of reference.

STAGE 14: BEGIN ASSEMBLING ROBI'S BODY

In this stage, you will assemble the first parts of Robi's body, by attaching the right arm built and wired over the previous stages to his right body cover and shoulder servo.

This stage sees you begin building up Robi's body for the first time, which you will do by fitting the previously assembled right arm onto servo ID=16, which you wired and programmed in the previous stage.

You will also use a servo horn for the first time. These fittings are used to connect a moving part, in this case Robi's right arm, to a servo's shaft. The part features a D-cut, and care must be taken to ensure the parts fit neatly.



YOUR PARTS

1 Right body cover
2 Servo horn (shoulder)
3 M2 x 4.5mm screws x 3 (one is a spare)

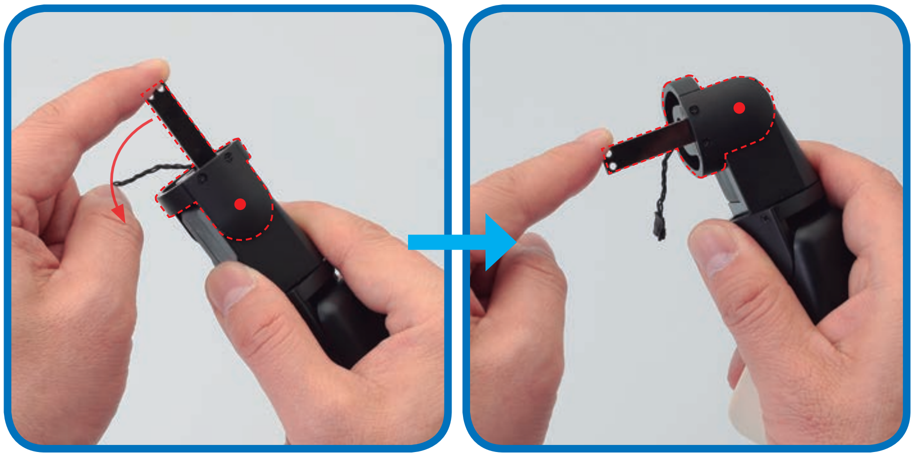
SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

Right arm assembly (Stage 12)

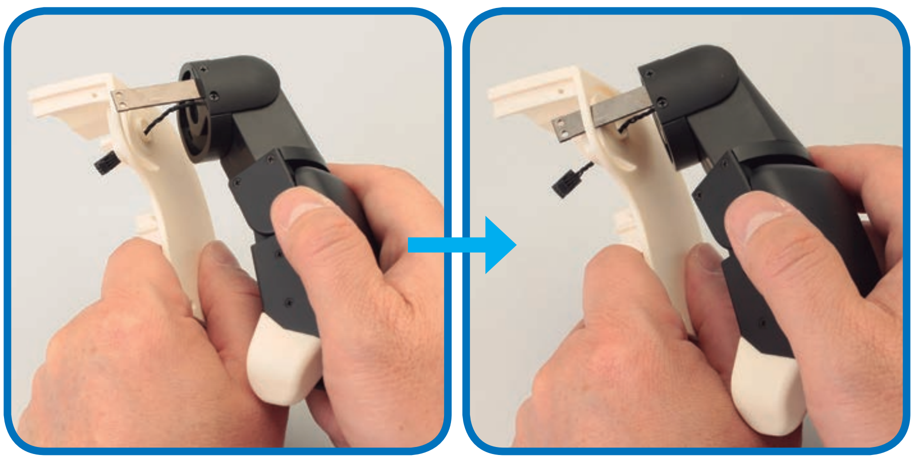
Wired servo (Stage 13)

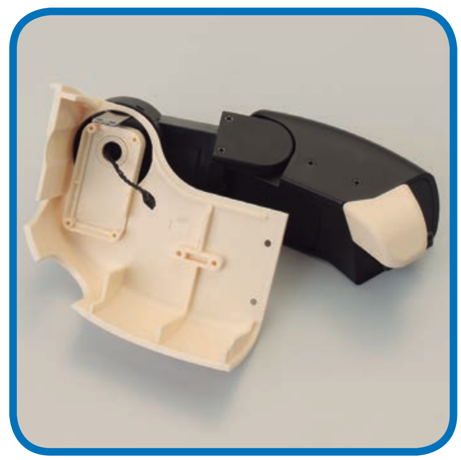
THE RIGHT ARM AND BODY COVER



1 Carefully rotate the shoulder joint at the top of the right arm assembly so it sits at right angles to the rest of the arm.

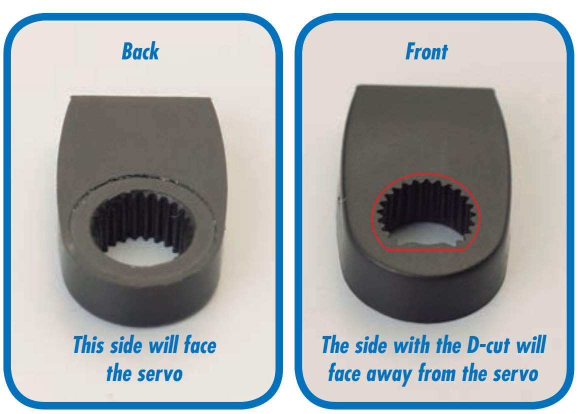
2 Line up the right arm to the right body cover.

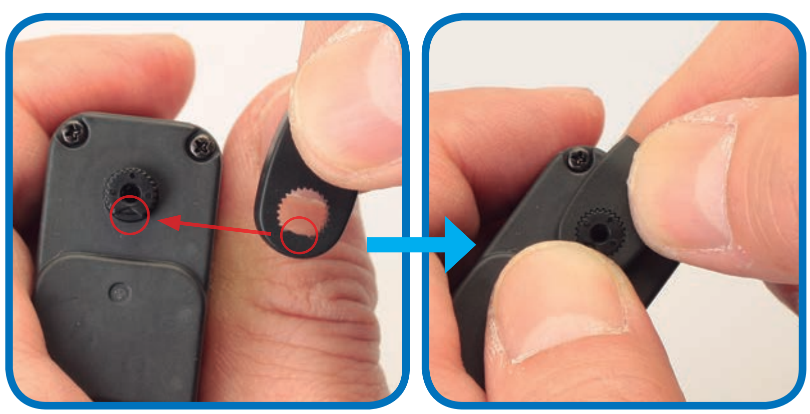




3 Fit the servo cable emerging from the top of the arm through the circular hole in the right body cover, and the metal shoulder joint through the C-shaped slot above it, as shown.

4 Join the parts so they sit together snugly and the cable and joint are fully inside the cover.



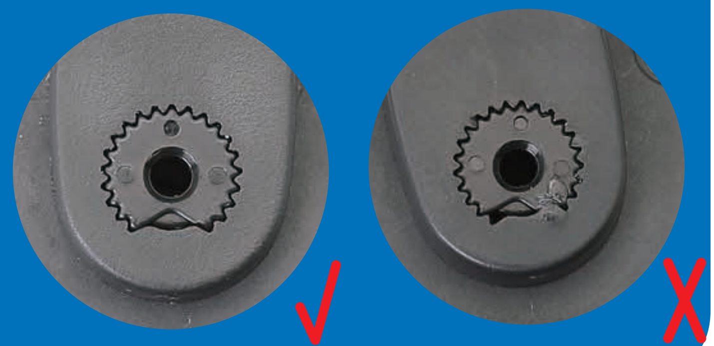


5 Inspect the servo horn: the hole in its front side has a D-cut profile, and will face outwards, while the back of the part has a round hole that will face the servo.

6 Find the servo from Stage 13 and line up the servo horn so that its front faces away from the servo. Then, with the servo horn perfectly straight with its D-cut section matching the cutaway on the servo's shaft, press the parts together. Do not force the parts, and see the Close Up box on the next page for more information.

THE D-CUT

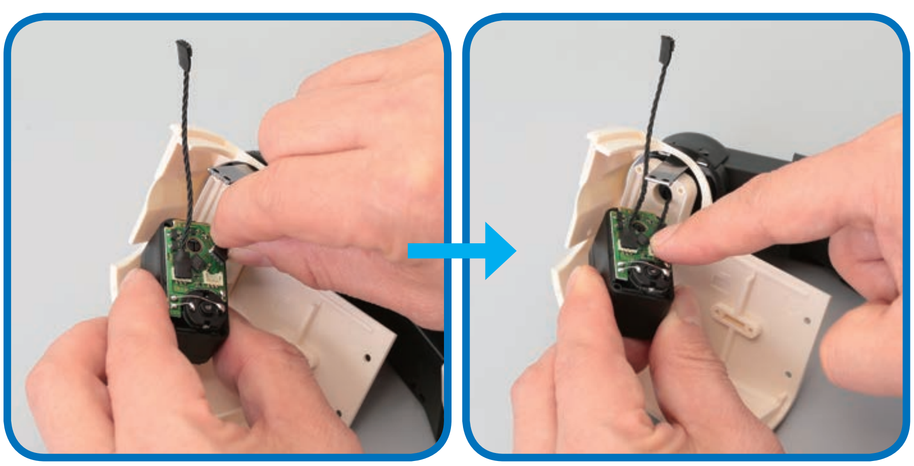
As explained in Stage 12, it is very important that you take care when working with the servos. Each joint, such as the servo horn in this stage, will have a specially moulded cutaway section known as the D-cut, which, when used properly, will fit snugly around the servo shaft to ensure it is set correctly. When these parts are aligned as they should be, it will be easy to slot them together. If it is in any way difficult to join the parts, remove them and check you have the D-cut section lined up to the shaft's cutaway and begin again. Never force the parts, as this could damage both of them.



FITTING THE RIGHT ARM SERVO

7 Remove the four long screws from the back of the servo to access the front panel.

8 Remove the front panel, being careful not to pull the servo cable away as you do so. You will not be using this panel again.

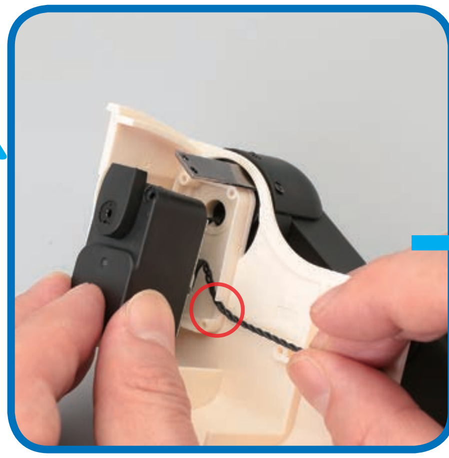


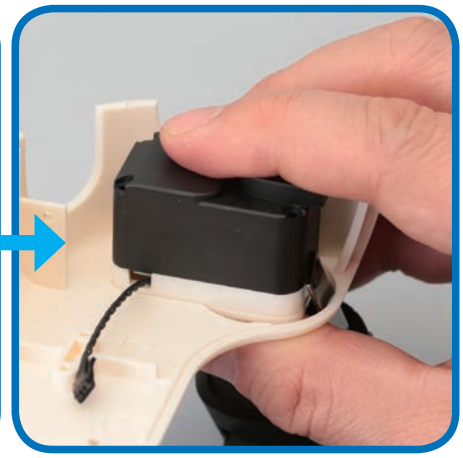
9 Hold the open servo close to the inside of the right body panel, then carefully press the free, padded connector of the right arm's servo cable into its free socket.

TIP!

This servo cable had protective pads fitted to its connectors, but if you find it difficult to attach with the pad in place, it is possible to apply it afterwards. To do this, very carefully fit the connector into the servo's socket, then remove a pad from the backing sheet (Stage 3) and place on top of the connector and press down to seal the part.

Before sealing the servo, the cables need to be arranged in such a way that they do not become tangled. To do this, bend the cable leading to the arm to the side of the servo's body, and the other back on itself and then out at right angles on the opposite side. Be careful not to disrupt either connector as you do this.





10 With the circled cable bent around to run into the notch on the inside of the body cover (see box to the left), press the servo onto the correspondingly shaped mount.

As you tighten the screws, check that the servo cable and metal shoulder joint are still in position on the outside of the body cover.

11 Hold the parts in place, then using the four screws removed in Step 7, fasten the servo to the body cover.

12 Remove the two screws on the shoulder joint and remove the protective panel.

13 Very gently pull the exposed portion of the servo cable out from the elongated hole in the metal joint. Do not pinch the cable.

Eliminating any slack

The aim of this step is to remove any slack in the servo cable. Any loose sections along the servo cables set within Robi's moving parts, in this case the shoulder joint, are at risk of rubbing, or getting caught or pinched. This can ultimately damage the cables, and even cause Robi to malfunction, so be sure to follow these steps carefully and ensure that all the cables are set appropriately within the joints.

14 Lift the cable away from the side of the body panel so there is no slack, then press down with your finger over the circular hole in the arm to prevent the cable running back into the body.

15 Keeping the cable tight to the body side, bunch the slack up into a circle above the hole into the upper arm frame, to match the shaped recess in the inside of the shoulder panel removed in Step 12.

16 Fit the bunch of cable slack inside the recess, then, making sure not to pinch any part of the cable, replace the shoulder panel and tighten with its original screws.

17 Line up the two holes at the end of the metal shoulder joint with those on the servo horn.

18 Insert and tighten an M2 x 4.5mm screw into each hole to secure the parts.

THE SERVO SCREWS

CLOSE-UP

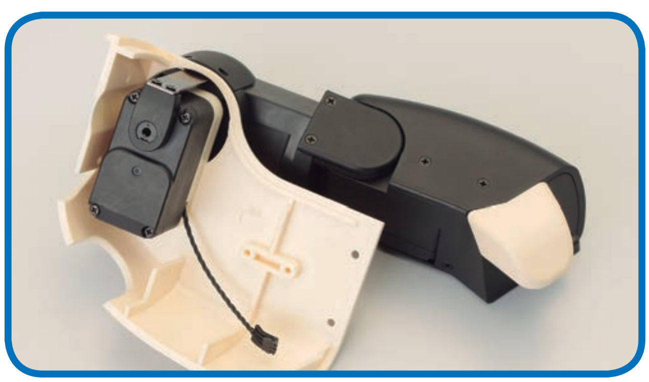
It is vitally important that the servo that moves Robi's shoulder is not hampered in any way, as this may cause a malfunction. Therefore, be sure to check that the screws on either side of the servo horn that connects to Robi's shoulder joint are fully inserted and do not obstruct the part's movement.

The servo horn will move back and forth, over the screws circled in the photos to the left. Double-check that these are tightened fully into their holes so as to allow the servo horn freedom of movement.

To check that the servo horn is able to move easily, hold the body panel in one hand and with the other, slowly rotate Robi's right arm so that the servo horn covers each of the screws. If you feel any resistance, tighten the screw further and repeat until the arm is able to move freely.

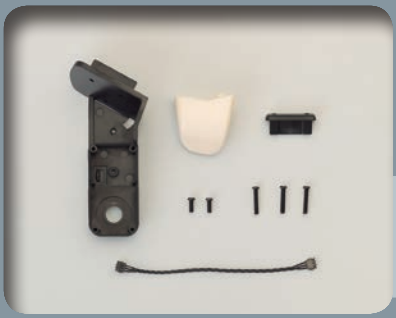
Assembled right arm and body cover

The first join between one of Robi's limbs and his body has been made, with the servos used to move Robi's right arm wired into the one hidden within his body to control his shoulder movements. Store your assembly away safely until next time.



COMING IN PACK 5

Your next **FOUR** complete Stages, as Robi starts to come to life!

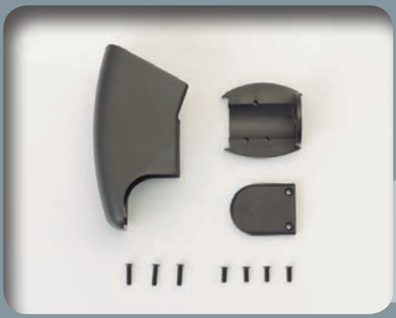


Stage 15
Start building Robi's left forearm

PARTS PROVIDED
The first parts of Robi's left arm, and another servo cable.

Stage 16
Set the ID number of Robi's left forearm servo

PARTS PROVIDED
The servo that will operate Robi's left forearm.



Stage 17
Complete Robi's left forearm

PARTS PROVIDED
The left forearm cover, and the elbow frame and panel.

Stage 18
Begin building Robi's left upper arm

PARTS PROVIDED
The left upper arm frame, elbow panel and a servo cable.